Cost Control and Group Behavior

Lecture 4

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Introduction

Cost control is equally important to all companies, regardless of size.

Small companies generally have tighter monetary controls because the failure of even one project can put the company at risk, but they have less sophisticated control techniques.

Large companies may have the luxury to spread project losses over several projects, whereas the small company may have few projects.

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Introduction

Cost control implies good cost management, which must include:

- Cost estimating
- Cost accounting
- Project cash flow
- Company cash flow
- Direct labor costing
- Overhead rate costing
- Other tactics, such as incentives, penalties, and profit-sharing

Cost control is actually a subsystem of the management cost and control system (MCCS) rather than a complete system per se.

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Understanding Control

The requirements for an effective control system (for both cost and schedule/performance) should include:

- Thorough planning of the work to be performed to complete the project
- Good estimating of time, labor, and costs
- Clear communication of the scope of required tasks
- A disciplined budget and authorization of expenditures
- Timely accounting of physical progress and cost expenditures
- Periodic reestimation of time and cost to complete remaining work
- Frequent, periodic comparison of actual progress and expenditures to schedules and budgets, both at the time of comparison and at project completion

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Understanding Control

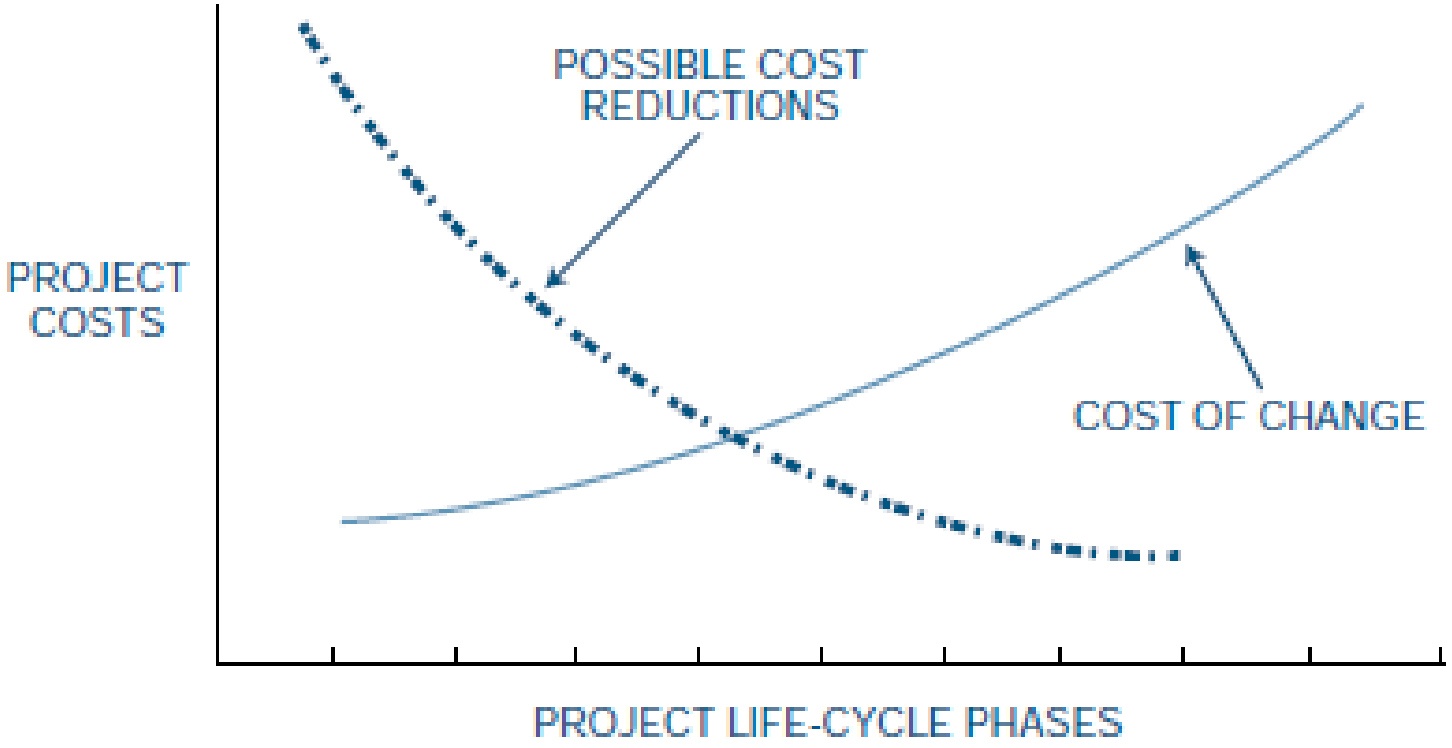


Fig. 1. Cost reduction analysis

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Understanding Control

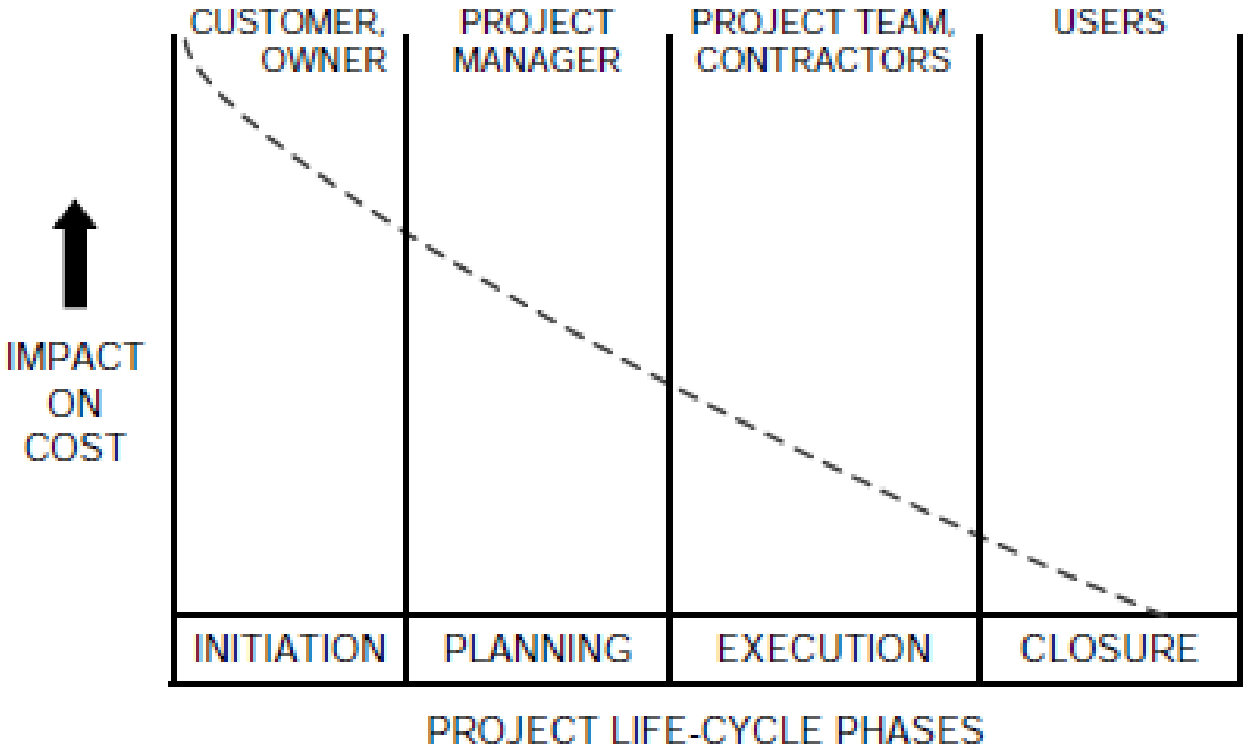


Fig. 2. People with the ability to influence cost

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The Operating Cycle

The management cost and control system (MCCS) takes on paramount importance during the operating cycle of the project. The operating cycle is composed of four phases:

- Work authorization and release (phase II)
- Cost data collection and reporting (phase III)
- Cost analysis (phase IV)
- Reporting: customer and management (phase V)

These four phases, when combined with the planning cycle (phase I), constitute a closed system network that forms the basis for the management cost and control system.

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Budgets

All budgets must be traceable through the budget “log,” which includes:

- Distributed budget
- Management reserve
- Undistributed budget
- Contract changes

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Budgets

In addition to the “normal” performance budget and the management reserve budget, there are two other budgets:

● Undistributed budget, which is that budget associated with contract changes where time constraints prevent the necessary planning to incorporate the change into the performance budget. (This effort may be time-constrained.)

● Unallocated budget, which represents a logical grouping of contract tasks that have not yet been identified and/or authorized.

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The Earned Value Measurement System (EVMS)

Earned value management (EVM) is a systematic process that uses earned value as the primary tool for integrating cost, schedule, technical performance management, and risk management.

The EVMS emphasizes prevention over cures by identifying and resolving problems early. The EVMS is an early warning system allowing for early identification of trends and variances from the plan. The EVMS provides an early warning system, thus allowing the project manager sufficient time to make course corrections in small increments. It is usually easier to correct small variances as opposed to large variances.

Therefore, the EVMS should be used continuously throughout the project in order to detect the variances while they are small and possibly easy to correct. Large variances are more difficult to correct and run the risk that the cost to correct the large variance may displease management to the point where the project may be canceled.

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Variance and Earned Value

● Budgeted cost for work scheduled (BCWS) is the budgeted amount of cost for work scheduled to be accomplished plus the amount or level of effort or apportioned effort scheduled to be accomplished in a given time period.

● Budget cost for work performed (BCWP) is the budgeted amount of cost for completed work, plus budgeted for level of effort or apportioned effort activity completed within a given time period. This is sometimes referred to as “earned value.”

● Actual cost for work performed (ACWP) is the amount reported as actually expended in completing the work accomplished within a given time period.

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Variance and Earned Value

- Cost variance (CV) calculation:

$$CV = BCWP - ACWP$$

A negative variance indicates a cost-overrun condition.

- Schedule variance (SV) calculation:

$$SV = BCWP - BCWS$$

A negative variance indicates a behind-schedule condition.

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Variance and Earned Value

In the analysis of both cost and schedule, costs are used as the lowest common denominator. In other words, the schedule variance is given as a function of cost. To alleviate this problem, the variances are usually converted to percentages:

Cost variance % (CVP) = CV / BCWP

Schedule variance % (SVP) = SV / BCWS

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Variance and Earned Value

It is important to understand the physical meaning of CV and SV. Consider the following example:

- BCWS = $1000; BCWP = $800; ACWP = $700

In this example, the units are dollars. The units could have just as easily been hours, days, or weeks.

In this example, CV = $800 - $700 = $100. Because CV is a positive value, it indicates that physical progress was accomplished at a lower cost than the forecasted cost. This is a favorable situation. Had CV been negative, it would have indicated that physical progress was accomplished at a greater cost than what was forecasted. If CV = 0, then the physical accomplishment was as budgeted.

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Variance and Earned Value

Although CV is measured in hours or dollars, it is actually a measurement of the efficiency with which physical progress was accomplished compared with the plan. To correct a negative cost variance, emphasis should be placed upon the productivity rate (i.e., burn rate) at which work is being performed.

Returning to the above example, SV = $800 - $1000 = - $200. In this example, the schedule variance is a negative value, indicating that physical progress is being accomplished at a slower rate than planned. This is an unfavorable condition. If the schedule variance were positive, this would indicate physical progress being accomplished at a faster rate than planned. If SV = 0, physical progress is being accomplished as planned.

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Variance and Earned Value

In addition to calculating the cost and schedule variances in terms of dollars or percentages, we also want to know how efficiently the work has been accomplished. The formulas used to calculate the performance efficiency as a percentage of EV are:

Cost performance index (CPI) = BCWP / ACWP

Schedule performance index (SPI) = BCWP / BCWS

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Variance and Earned Value

If CPI = 1.0, we have perfect cost performance. If CPI < 1.0, physical progress is being accomplished at a greater cost than forecasted. This is unfavorable. If CPI > 1.0, physical progress is being accomplished at less than the forecasted cost, which is favorable.

Similar to CV, CPI measures the efficiency by which the physical progress was accomplished compared to the plan or baseline. For an unfavorable value of CPI, emphasis should be placed upon improving the productivity by which work was being performed.

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Variance and Earned Value

If SPI = 1.0, we have perfect schedule performance. If SPI < 1.0, physical progress was accomplished at a slower rate than what was planned. This is unfavorable.

If SPI > 1.0, physical progress was accomplished at a faster rate than what was planned, which is favorable. For an unfavorable value of SPI, emphasis should be placed upon improving the timeliness of the physical progress.

SPI and CPI are expressed as ratios compared to the performance factor of 1.0 whereas CV and SV are expressed in hours or dollars.

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Variance and Earned Value

The difficulty in performing variance analysis is the calculation of BCWP because one must predict the percent complete. The simplest formula for calculating BCWP is:

$$BCWP = (\% \text{ complete}) * BAC$$

To complete our analysis of the status of a project, we must determine the budget at completion (BAC) and the estimate at completion (EAC).

We can calculate the variance at completion (VAC):

$$VAC = BAC - EAC$$

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Understanding Control

Question	Answer	Acronym
How much work *should* be done?	Budgeted cost of work scheduled	BCWS
How much work *is* done?	Budgeted cost of work performed	BCWP
How much *did* the "is done" work cost?	Actual cost of work performed (actuals)	ACWP
What was the total job *supposed* to cost?	Budget at completion (total budget)	BAC
What do we *now* expect the total job to cost?	Estimate at completion or latest revised estimate	EAC LRE

Table 1. The Parameters For Variance Analysis

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Variance and Earned Value

Using the above definitions, we can calculate the variance at completion (VAC):

$$VAC = BAC – EAC$$

The estimated cost-to-complete (EAC) ratio:

$$EAC = (ACWP/BCWP) \text{ X } BAC = BAC/CPI$$

New project length = original project length / SPI

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The Meaning of a Group

Group is defined as two or more employees who interact with each other in such a manner that the behavior and/or performance of a member is influenced by the behavior and/or performance of other members.

Formal groups are groups created by managerial decision to accomplish stated goals of the organization.

The demands and processes of the organization lead to the formation of two types of formal groups: command and task.

Informal groups are groups that arise from individual efforts and develop around common interests and friendships rather than deliberate design.

Two specific informal groups exist: interest and friendship.

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Why People Form Groups

Formal and informal groups form for various reasons. Some reasons involve needs, proximity, attractions, goals, and economics.

The Satisfaction of Needs

The desire for need satisfaction can be a strong motivating force leading to group formation. Specifically, some employees' security, social, esteem, and self-actualization needs can be satisfied to a degree by their affiliation with groups.

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Stages of Development

The Five-Stage Model

This model explains the group development process in terms of a maturity cycle. The five stages are referred to as follows:

1. Forming. This is the breaking the ice, getting acquainted stage. Group members are trying out behaviors, testing their position, and asking questions of other group members. The group is establishing loose, but specific, ground rules.

2. Storming. This stage is characterized by conflict because members are arguing, debating, and experimenting with roles, advice offered by other members, and attempts to move into leadership roles. The group's hierarchy starts to take shape.

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Stages of Development

3. Norming. The group starts to work more effectively together. There is a sense of togetherness or attraction to being a part of a group. This is the beginning of cohesiveness. A set of group-driven expectations is communicated within the group.

4. Performing. A group structure, hierarchy, and norms are in place. The group is focused on accomplishing goals and being an efficient unit. This is an important stage in that the group is mature.

5. Adjourning. The group prepares to disband. The goals have been accomplished and tasks finished. Some members will be saddened over the loss of member contact and cohesiveness in the future.

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Stages of Development

The Punctuated Equilibrium Model

Some believe that the five-stage model is too static and unrealistic. The punctuated equilibrium model (PEM) presents group development as a three-phase concept. The first phase occurs when groups define tasks, establish goals, and consider various ways to execute the steps of the plan or project.

Somewhere along a time continuum, the group decides to change its perspectives and take action. The group reaches its own point of inertia or equilibrium. The group reaches the point of equilibrium when it realizes that time is running out. After this realization phase, there is then a final phase to accomplish tasks.

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Summary of Key Points

• A group consists of employees who interact in such a manner that the behavior or performance of one group member is influenced by the behavior or performance of other group members.

• By being aware of group characteristics and behaviors, managers can be prepared for the potential positive and negative results of group activities. A manager can proactively intervene to modify the perceptions, attitudes, and motivations that influence the results.

• People are attracted to groups because of their potential for satisfying needs, their physical proximity and attraction, and the appeal of group goals and activities. In essence, people are attracted to one another; that is a natural process.

• Groups develop at different rates and with unique patterns that depend on the task, the setting, the members' individual characteristics and behavior patterns, and the manager's style of managing.

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Summary of Key Points

• Characteristics of groups include structure, status hierarchy, roles, norms, leadership, cohesiveness, and intergroup conflict. These characteristics pervade all groups. In an informal group, they emerge from within the unit; in a formal group, they are established by the managerial process.

• Group characteristics provide a degree of predictability for the members that is important to the group and to the outside (e.g., management, other groups). An unstable or unpredictable group is a problem for its members and for others who interact with it.

• Each group possesses some degree of cohesiveness. This attractiveness of the group can be a powerful force in influencing individual behavior and performance.

• Research studies indicate that cohesive groups can formulate goals and norms that may not agree with those of management. When a group's goals and norms are incongruent with the organization's, some form of managerial intervention is necessary.

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Summary of Key Points

• Groupthink is the desire among cohesive group members to be in agreement, which supersedes considering solutions to problems or situations that are controversial or in which unanimous agreement is not possible.

• Teams are a special type of task group. Three common categories of teams exist in organizations: (1) problem-solving teams, (2) cross-functional teams, and (3) self-directed work teams.

• Managers form teams predominantly to enhance productivity. Teams also form due to flattening organizations, need for flexibility and quick decisions, workforce diversity, improved quality, and increased customer satisfaction.

• The requirements for effective teams include top-level commitment and provision of clear goals; management–employee trust; willingness to take risks and share information; and time, resources, and a commitment to training.

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Summary of Key Points

• The concept of role is vital to an understanding of group behavior. A role consists of the expected behavior patterns attributed to a particular position. Most individuals perform multiple roles, each with its own role set (others' expectations for the role). An individual involved in many different roles, each having a complex role set, faces the ultimate in complexity of individual behavior.

• In organizations, there may be as many as three perceptions of the same role: the organization's, the group's, and the individual's. When an individual faces two or more simultaneous role requirements for which the performance of one precludes performance of the other(s), she experiences role conflict.

• Three different types of role conflict—person–role, intrarole, and interrole—can occur in organizational settings. Research has shown that consequences of role conflict to Management can minimize certain types of role conflicts and should be continually aware that the consequences of conflict to the organization can include ineffective performance by individuals and groups.

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Reference

Project Management: A Systems Approach to Planning, Scheduling, and Controlling, 12th ed. Harold Kerzner, ISBN-10: 9781119165354, ISBN-13: 978-1119165354, 2017

Organizations: Behavior, Structure, Processes, 14th edition, Gibson J. et al., ISBN 978-0-07-811266-9, MHID 0-07-811266-4, McGraw-Hill, 2012

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